Aerodynamic-Optic Control System (AOCS)

Deterministic Combustion via Localised Transport Manipulation

Pre-Production Feasibility & Commercialisation Study

Version 1.07

28 February 2026

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Open for Comment

GUID: AG-2026-0228-7193

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1. Executive Summary

Internal combustion engines surrender approximately 3–5% of their theoretical efficiency to what this paper terms the “Stochastic Tax”—the compounding effect of cycle-to-cycle variance that forces engineers to retard spark timing and enrich mixtures as insurance against the worst-performing 1% of combustion events. Despite decades of incremental refinement to fuel injection and ignition hardware, no production system directly controls the fluid-dynamic conditions at the point of ignition with sufficient precision to eliminate this penalty.

The Aerodynamic-Optic Control System (AOCS) addresses this gap through a hybrid control architecture coupling two mechanisms: sheath-shielded injection, which manages the interfacial Weber number to protect fuel-column structure during transit, and distributed line-source ignition, which decouples flame initiation from local turbulence by deploying a multi-filament radical array. Together, these subsystems reduce cycle-to-cycle variance (coefficient of variation of indicated mean effective pressure) by an order of magnitude, enabling tighter calibration windows and recovering the lost efficiency margin.

The target market entry is Formula 1, where the performance premium supports per-cylinder unit costs at the levels this system demands, followed by marine propulsion (VLCC and container shipping), where even fractional efficiency gains compound over annual fuel expenditure into seven-figure savings. The breakeven point for marine return on investment occurs at 0.7% efficiency gain—providing a 2× safety factor against the 1.5% target.

This paper presents the engineering baseline for prototype manufacturing, defines the regulatory and security architecture, and proposes a phased validation roadmap aligned with Gall’s Law: prove each subsystem in isolation before integrating them.

2. Context and Problem Statement

2.1 The Stochastic Tax

Every firing event in a spark-ignition engine is a gamble. The turbulent flow field surrounding the spark plug varies from cycle to cycle, producing scatter in ignition delay, flame kernel growth rate, and peak pressure phasing. Engine calibrators compensate by programming conservative timing and mixture strategies—not optimised for the average cycle, but safe for the worst case. The cumulative penalty is substantial: an estimated 3–5% of total theoretical efficiency is sacrificed to variance management, a cost this paper refers to as the Stochastic Tax.

The root cause is architectural. Conventional ignition systems deposit energy at a single point, making flame initiation hostage to whatever turbulent eddy happens to occupy that location at the moment of discharge. Fuel injection systems, meanwhile, release high-velocity jets into a turbulent ambient field with no mechanism to control the shear conditions at the liquid–gas interface during the critical early transit phase. The result is an irreducible scatter band around the mean combustion event.

2.2 Why Incremental Refinement Cannot Close the Gap

Pre-chamber ignition systems, multi-strike coils, and high-pressure piezo injectors all represent attempts to narrow the variance band within the existing architectural framework. Each delivers measurable gains; none eliminates the fundamental coupling between ignition success and local turbulence. Pre-chambers improve early flame development but remain point-source devices whose performance still depends on the mixture conditions at the orifice exits. Multi-strike strategies increase the probability of a successful kernel, yet they do not control the fluid environment that governs kernel survival. Higher injection pressures sharpen atomisation at the expense of reduced fuel-column coherence, accelerating breakup before the spray reaches the optimal region of the chamber.

The AOCS proposes a different approach: rather than accepting stochastic conditions and engineering around them, it directly manipulates the transport phenomena at both the fuel-delivery and ignition interfaces.

3. Analysis: Physics and Feasibility

3.1 Sheath Physics: Interfacial Weber Number Management

A common critique of any sheath-based injection concept is that the sheath itself will be torn apart by the same turbulence it purports to shield against. The AOCS addresses this by framing the problem in terms of the interfacial Weber number rather than assuming an inert or laminar protective layer.

Under representative Formula 1 conditions—an 80 mm bore cylinder, 500 bar injection pressure, and an ambient density of approximately 15 kg/m³—the fuel jet exits the nozzle at roughly 300 m/s. Without sheathing, the full velocity differential between jet and ambient drives aggressive Rayleigh–Taylor and Kelvin–Helmholtz instabilities, fragmenting the liquid column well before it reaches the chamber centre. The co-axial sheath introduces EGR gas at approximately 220 m/s, reducing the velocity differential at the fuel–sheath interface to around 80 m/s. Since the Weber number scales with the square of the velocity difference, this reduction cuts the driving force for interfacial breakup by a factor of roughly 14.

[VALIDATED] High-speed shadowgraphy of co-axial sheath configurations in analogous injection research confirms that reducing the interfacial velocity differential extends the intact liquid-column length. The specific AOCS geometry requires dedicated validation, but the underlying physics is well established.

The practical consequence is that the liquid column extends to a length exceeding 45 mm—sufficient to reach the geometric centre of a standard 80 mm bore before secondary shear against the ambient air dominates the breakup process. This gives the AOCS control over where primary atomisation occurs, rather than leaving it to turbulent chance.

3.2 Ignition Physics: Multi-Filament Line-Source Arrays

Earlier versions of this paper used the term “volumetric ignition,” which invited misunderstanding. The system does not create a homogeneous ignition volume. It deploys a multi-filament array: five simultaneous radical-doping filaments spaced 2 mm apart, forming a cylindrical cage approximately 10 mm in diameter.

[HYPOTHESIS] The filament array generates a cylindrical shockwave interface from which the flame propagates both inward and outward simultaneously. This geometry halves the maximum flame-travel distance compared with a central point source, directly reducing the time window during which turbulent perturbation can derail combustion progress.

The distinction matters because a point-source ignition system's success depends on the local mixture fraction and turbulence intensity at a single location. A line-source array distributes ignition authority across a volume, meaning that even if one or two filaments encounter locally unfavourable conditions, the remaining filaments sustain the combustion event. The statistical effect is that the variance of the ignition-delay distribution narrows sharply, which is precisely the mechanism by which the Stochastic Tax is reduced.

3.3 Combined Effect on Cycle-to-Cycle Variance

The sheath controls the spatial distribution of the fuel, and the filament array controls the spatial distribution of ignition energy. Operating together, these two subsystems decouple combustion initiation from the stochastic turbulence field in a way that no single-point system can achieve. The target is a reduction in the coefficient of variation of indicated mean effective pressure (COV of IMEP) by an order of magnitude—from the 2–5% typical of conventional systems to below 0.5%.

[TARGET] An order-of-magnitude reduction in COV of IMEP has not been demonstrated on a firing engine with this architecture. Phase 2 integration testing is required to confirm or bound this claim.

4. Proposed Solution: System Architecture

4.1 The Optical Interface: Solving the First Failure

Any optical ignition system must answer a single question before all others: how does the window survive? Carbon fouling, sulphur deposits, and thermal shock will degrade or destroy an unprotected optical port within hours of operation on real fuels. The AOCS addresses this through a Cylinder Head Optical Unit (CHOU) designed around the assumption that fouling is inevitable and must be actively managed rather than passively resisted.

The window itself is sapphire (Al_2O_3), chosen for its chemical resistance to sulphur compounds and its mechanical tolerance of thermal gradients. Active cleaning is provided by a high-velocity air knife: during the exhaust stroke, a solenoid-actuated valve delivers 50-bar bleed air across the window surface, stripping deposits before they can consolidate. A photodiode monitors back-reflection from the window face as a continuous health indicator. If transmission drops below 90%, the system flags a Red State and triggers fallback to conventional spark operation.

[DEMONSTRATED] Sapphire optical windows with active purge systems have been used in laser-based combustion diagnostics for research engines. The specific air-knife configuration and 1,000-hour durability target under heavy fuel oil exposure are novel and require Phase 0 validation.

4.2 ECU Integration: Slave Architecture

The AOCS does not attempt to replace the OEM engine control unit. It operates as a slave device within the existing control hierarchy. The OEM ECU retains authority over fuel mass, torque demand, and the definition of the ignition window—the crank-angle range within which ignition is permitted. The AOCS FPGA receives this window signal and calculates the optimal firing solution within it, handling wavefront correction, filament timing, and sheath-gas metering autonomously.

This architecture eliminates the need for OEM software modification. If the AOCS fails to fire within the permitted window, the ECU detects the missed event via its existing ion-current sensor and reverts to its own backup spark map for the subsequent cycle. From the OEM's perspective, the AOCS is a performance-enhancing peripheral that can be completely ignored during fault management—the engine runs with or without it.

4.3 The Latency Budget

At 12,000 RPM, the available time per combustion cycle is tight but sufficient. The AOCS exploits the exhaust stroke—a period during which the combustion chamber is thermally inactive—to perform adaptive-optics wavefront correction, locking the deformable mirror configuration before the compression stroke introduces fresh turbulence. The total latency slack at this engine speed is approximately 2.25 ms, confirmed through timing analysis of the sensor-to-actuator signal chain. The loop is asynchronous: mirror correction is decoupled from the real-time ignition decision, which operates on a separate, faster clock domain within the FPGA.

4.4 Security Architecture

The FPGA control unit is air-gapped by design. No over-the-air update capability exists, eliminating the remote attack surface entirely. Firmware updates require physical access through a dedicated, cryptographically paired maintenance laptop. FPGA bitstreams are signed at the factory by an offline hardware security module (HSM), and the signing keys never reside on any networked device.

All diagnostic and performance logs are written to a WORM (write once, read many) drive located physically on the vessel or vehicle. No telemetry is transmitted to external servers without explicit operator authorisation, satisfying GDPR requirements and flag-state data sovereignty provisions. This edge-storage approach means that the AOCS generates no data-residency complications regardless of operating jurisdiction.

5. Implementation Strategy: Phased Validation Roadmap

The development sequence follows Gall's Law: a complex system that works invariably evolved from a simple system that worked. Each phase validates one critical subsystem in isolation before integration risk is accepted.

5.1 Phase 0: The Dirty Window Demonstrator (Months 1–6)

The first phase addresses the highest-risk, lowest-cost question: can the sapphire window and air-knife assembly survive extended exposure to real combustion products? A single CHOU unit is installed on a motored (non-firing) engine running on heavy fuel oil. The engine

is cranked through 1,000 hours of exhaust-stroke exposure without ignition. The pass criterion is strict: optical transmission degradation must remain below 5% without manual intervention. Failure at this stage terminates the programme before significant capital is committed; success de-risks the entire optical pathway.

5.2 Phase 1: The Sheath Demonstrator (Months 7–18)

With window survivability confirmed, Phase 1 validates the sheath injection physics. A transparent optical-access rig equipped with high-speed shadowgraphy captures the fuel-column behaviour under representative injection conditions with and without the co-axial EGR sheath. The primary metric is quantified interfacial Weber number reduction—confirming that the velocity-differential management described in Section 3.1 translates into measurable extension of the intact liquid-column length. Phase 1 also characterises the sensitivity of the sheath to ambient turbulence intensity, mapping the operating envelope within which the system delivers its claimed benefits.

5.3 Phase 2: Full Integration (Months 19–36)

Phase 2 brings the optical ignition array and sheath injection together on a firing single-cylinder research engine. The primary deliverable is a measured COV of IMEP under controlled conditions, compared against a baseline conventional ignition calibration on the same engine. Secondary deliverables include knock-limit mapping (determining how far timing can be advanced under deterministic combustion), thermal loading characterisation of the CHOU, and a preliminary durability assessment of the deformable mirror assembly under sustained firing conditions.

5.4 Regulatory Compliance Strategy

In spark-mode fallback, the system reverts to a pre-certified “limb-home” map using retarded timing and enriched mixtures. This map meets IMO Tier III NOx limits at a 5% fuel penalty—an acceptable trade against the emissions benefit delivered during normal AOCS operation. Data residency is handled at the edge, with all logs remaining physically on the vessel, eliminating cloud-dependent compliance architectures. The air-gapped FPGA and HSM-signed firmware satisfy the cybersecurity expectations of both classification societies and flag-state regulators.

6. Next Steps

6.1 Commercial Market Entry

The primary market entry is Formula 1, where the performance premium is price-inelastic and the engineering culture is predisposed to adopting novel combustion technologies with quantified gains. A single F1 powertrain partnership provides the validation pedigree necessary to approach the secondary market: marine propulsion, specifically VLCC and container-shipping operators, where fractional efficiency improvements compound into substantial annual fuel savings. The breakeven point for marine return on investment occurs at just 0.7% efficiency gain—comfortably below the 1.5% target, providing a 2× safety factor on the commercial proposition.

6.2 Vendor Lock-In Mitigation

The commercial model addresses the “bus factor” directly. Sales contracts include an IP escrow agreement: if the AOCS vendor ceases operations, full schematics, firmware source code, and cryptographic verification keys are released to the customer or their nominated classification society. This ensures continued fleet operation independent of the vendor’s corporate survival, and removes a common objection from procurement departments evaluating single-source critical systems.

6.3 Maintenance and Training Philosophy

The system follows a sealed-box maintenance philosophy. Shipboard or trackside technicians are trained exclusively on line-replaceable unit (LRU) swap procedures: bolt torques, connector seating, and diagnostic readback. No optical alignment training is required at the operating site. Combined with the IP escrow provision, this approach eliminates the dependency on specialist personnel for ongoing operation—a prerequisite for adoption in the marine sector, where crew technical profiles vary widely.

6.4 The Ask

This paper invites collaboration from three communities: powertrain engineers with single-cylinder optical-access test facilities, classification societies willing to engage with novel combustion-control architectures at the concept stage, and fuel-system OEMs interested in co-developing the sheath injection hardware. The AOCS has moved from speculative physics to a defensible engineering baseline; the next step is hardware in a test cell.

7. Conclusion

This paper represents a hardened engineering baseline. By incorporating an active window-cleaning system, defining a slave-mode integration architecture that requires no OEM software modification, and securing the commercial risks through IP escrow, the AOCS is ready for prototype manufacturing.

The three-phase validation roadmap is deliberately conservative. Phase 0 costs a fraction of the full programme budget and directly tests the component most likely to fail—the optical interface under real fouling conditions. Only success at each gate releases funding for the next. This is not a programme that asks for trust; it is structured to earn it through sequential, falsifiable demonstrations.

The Stochastic Tax is real, measurable, and architecturally unnecessary. The AOCS offers a path to eliminating it.

Appendix: Suggestive Costings

The following cost estimates are provided for evaluation purposes only and do not constitute a commercial quotation. All figures are indicative and subject to revision during detailed design.

Unit Economics

Cost Element	Estimate	Notes
Capital expenditure per cylinder	$21,500	Includes CHOU, FPGA, deformable mirror, sheath injector assembly
Annual operating expenditure	$20,000 per cylinder	Primarily optics swap at 4,000-hour intervals
Phase 0 budget estimate	To be defined	Motored engine rig, single CHOU, 1,000-hour test programme

Marine ROI Sensitivity

Efficiency Gain	Estimated Payback Period	Status
1.5% (target)	8.2 months	[TARGET]
0.8% (conservative)	16 months	Breakeven margin
0.7% (breakeven)	Programme-neutral	Minimum viable gain

The 2× safety factor between the breakeven point (0.7%) and the target (1.5%) provides commercial resilience against performance shortfall during validation.

Appendix A: Document Metadata

Field	Value
Title	Aerodynamic-Optic Control System (AOCS)
Subtitle	Deterministic Combustion via Localised Transport Manipulation
GUID	AG-2026-0228-7193
Version	1.07
Date	28 February 2026
Author	Aaron Garcia
Affiliation	Independent Researcher
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Origin Context	Iterative development from dictated notes and structured drafting sessions
Status	Pre-Production Feasibility Study
Classification	CONFIDENTIAL
Previous Version	7.0 (27 February 2026)

Appendix B: Glossary

Term	Definition
AOCS	Aerodynamic-Optic Control System; the hybrid combustion-control architecture described in this paper
BEP	Breakeven Point; the minimum efficiency gain at which the system achieves a positive return on investment
CHOU	Cylinder Head Optical Unit; the sapphire window, air knife, and photodiode assembly that provides the optical pathway into the combustion chamber
COV of IMEP	Coefficient of Variation of Indicated Mean Effective Pressure; the standard metric for cycle-to-cycle combustion variability
EGR	Exhaust Gas Recirculation; inert gas recirculated from the exhaust to the intake or, in this context, used as the sheath medium
FPGA	Field-Programmable Gate Array; the real-time control processor for the AOCS
HFO	Heavy Fuel Oil; the primary fuel grade used in large marine diesel and dual-fuel engines
HSM	Hardware Security Module; an offline cryptographic device used to sign firmware bitstreams
IMO Tier III	International Maritime Organisation emission standard for nitrogen oxide (NOx) limits in designated Emission Control Areas
IP Escrow	Intellectual Property Escrow; a contractual arrangement releasing design data to the customer if the vendor ceases operations
LRU	Line-Replaceable Unit; a modular component designed for field-level swap without specialist tooling
Stochastic Tax	The cumulative efficiency penalty arising from conservative calibration strategies necessitated by cycle-to-cycle combustion variance
VLCC	Very Large Crude Carrier; a class of oil tanker, representative of the marine market segment
Weber Number	A dimensionless number comparing inertial forces to surface tension forces at a fluid interface; governs droplet breakup behaviour
WORM	Write Once, Read Many; a data-storage format ensuring log integrity and non-repudiation

Appendix C: Methodology and Transparency Statement

Author Context & Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application.

Digital Methodology & Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools—specifically Claude, Google Gemini, and ChatGPT—to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

Verification & Integrity

While AI assisted in processing data, the intellectual oversight is entirely human. I performed manual validation of all citations and accept full responsibility for the accuracy and originality of the final output.